
Open-Set Wafer Defect Classification via Diagonal-Covariance Energy–Mahalanobis Fusion

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Abstract

Semiconductor wafer inspection systems are typically trained under a closed-set assumption, where every test sample is assumed to belong to one of the known training classes. In practice, however, previously unseen defect patterns may emerge due to process variations, contamination, or equipment failures. Standard classifiers often assign such samples to known classes with high confidence, resulting in unreliable predictions. In this work, we investigate open-set wafer defect classification using a ResNet18 backbone combined with post-hoc uncertainty estimation and feature-space out-of-distribution detection. We evaluate Maximum Softmax Probability (MSP), Energy-based detection, and Mahalanobis distance. Furthermore, we propose a diagonal-covariance Energy–Mahalanobis soft fusion strategy designed for small-data settings. Experimental results show that the proposed method improves macro F1 from 0.517 to 0.870, corresponding to a 68.4% relative improvement over the closed-set baseline.

1 Introduction

Wafer inspection is a critical component of semiconductor manufacturing because microscopic defects can significantly affect production yield and reliability. Although deep neural networks achieve strong performance on supervised defect classification tasks [1], they are typically trained under a closed-set assumption.

In practical manufacturing environments, previously unseen defect patterns may emerge after deployment. Conventional classifiers tend to assign such samples to known classes with high confidence, resulting in unreliable predictions.

To address this limitation, we formulate wafer inspection as an open-set recognition problem. Instead of forcing every sample into a known class, the model should be able to reject samples that deviate from the training distribution.

Our contributions are:

- Evaluation of MSP, Energy-based detection, and Mahalanobis distance for wafer open-set recognition.
- Investigation of covariance calibration under a small-data regime.
- Proposal of a diagonal-covariance Energy–Mahalanobis soft fusion method.

2 Dataset

The dataset consists of three known classes (DIE_BROKEN, NORMAL, NO_DIE) and two unseen classes (DIE_CRACK, DIE_INK). The classifier is trained only on the known classes, while unseen classes are

introduced exclusively during evaluation. The dataset contains 109 training samples, 75 validation samples, and 75 test samples.

3 Method

We employ an ImageNet-pretrained ResNet18 backbone [1] and train a classifier over only the three known classes using cross-entropy loss. The classifier outputs

$$f(x) \in \mathbb{R}^3$$

and does not contain an explicit unknown class. Although the resulting model achieves strong closed-set performance, it is unable to explicitly reject previously unseen defect categories.

3.1 Why Open-Set Recognition is Needed

To understand the limitations of closed-set classification, we visualize Grad-CAM activation maps in Figure 1. For known classes, the network focuses on semantically meaningful defect regions. In contrast, unseen defects often activate regions similar to visually related known classes. For example, the DIE_CRACK sample produces an activation pattern similar to NORMAL, while DIE_INK activates regions similar to DIE_BROKEN. Consequently, the classifier assigns unseen defects to known classes with high confidence, motivating the need for explicit open-set rejection mechanisms.

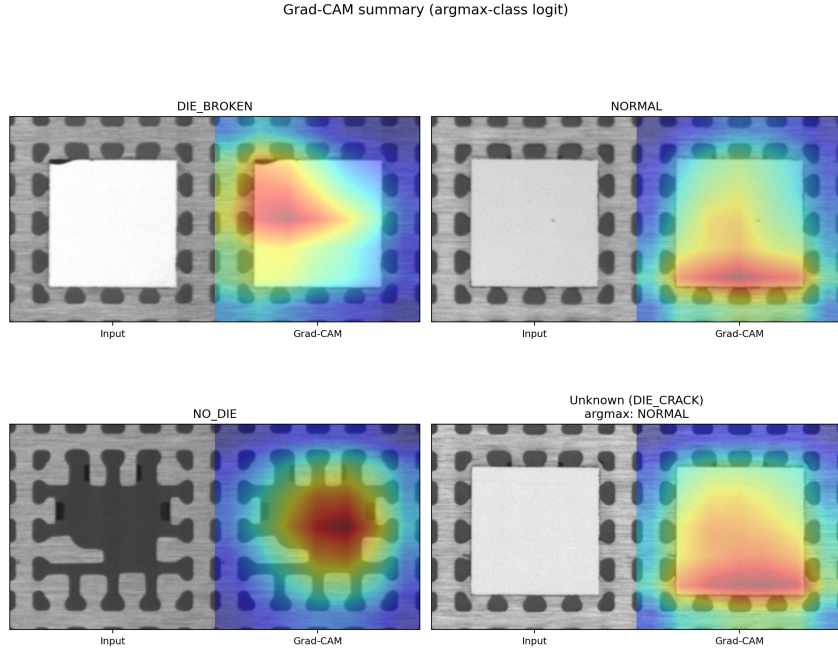


Figure 1: Grad-CAM visualization for known and unseen defect samples. Unknown defects often activate regions similar to known categories, explaining the failure of closed-set classification.

MSP. Following Hendrycks and Gimpel [2], Maximum Softmax Probability rejects a sample when

$$\max_k p(y = k|x)$$

falls below a threshold.

Energy. Following Liu et al. [3], Energy-based detection computes

$$E(x) = -T \log \sum_i \exp(f_i(x)/T),$$

where higher energy indicates a higher likelihood of belonging to an unseen class.

Table 1: Open-set classification performance on the test set.

Method	Known Acc	Unknown Prec	Unknown Recall	Macro F1
Argmax	0.971	0.000	0.000	0.517
MSP	0.914	0.727	0.200	0.607
Energy	0.800	0.696	0.400	0.632
Mahalanobis	0.971	1.000	0.600	0.790
Energy+Mahalanobis	0.971	1.000	0.625	0.801
Proposed	0.857	0.878	0.900	0.870

Mahalanobis Distance. Following Lee et al. [4], feature vectors are extracted from the penultimate layer and compared to class centroids using

$$D(x) = (z - \mu)^T \Sigma^{-1} (z - \mu).$$

3.2 Proposed Method

Our preliminary experiments revealed two important observations. First, Energy scores alone exhibited substantial overlap between known and unknown samples. Second, Mahalanobis distance provided significantly stronger separation between known and unknown samples. However, the training set contains only 109 samples, making full covariance estimation unstable.

We therefore replace the full covariance matrix with a diagonal approximation

$$\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_d^2).$$

In addition, Energy and Mahalanobis scores are standardized and combined through

$$s = \alpha E_z + (1 - \alpha) D_z.$$

This formulation combines uncertainty information from Energy with geometric information from Mahalanobis distance. The fusion coefficient was selected on the validation set, and the best configuration used $\alpha = 0.2$. The fusion coefficient α is selected on the validation set to maximize open-set macro F1. This improved separation directly translates into more reliable threshold selection and better open-set classification performance.

Figure 2 illustrates the score distributions of the three methods. The proposed fusion score produces the clearest separation between known and unknown samples.

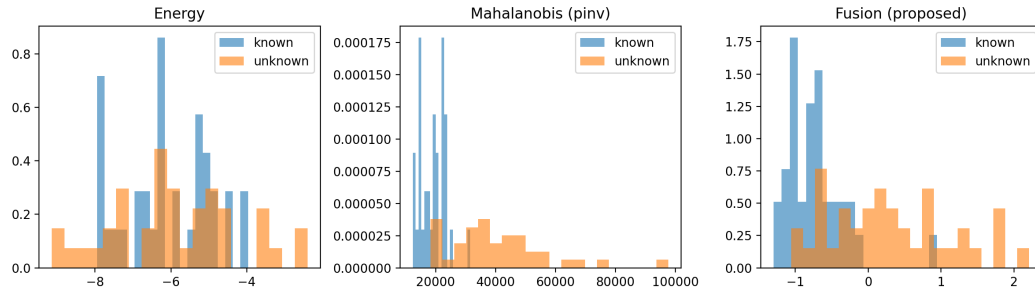


Figure 2: Energy, Mahalanobis, and fusion score distributions for known and unknown samples. The proposed fusion score provides the clearest separation.

4 Experimental Results

Table 1 summarizes the performance of all evaluated methods on the test split.

The closed-set baseline (Argmax) achieves high known-class accuracy but completely fails to reject unseen defects, resulting in an unknown recall of 0.0 and a macro F1 score of only 0.517.

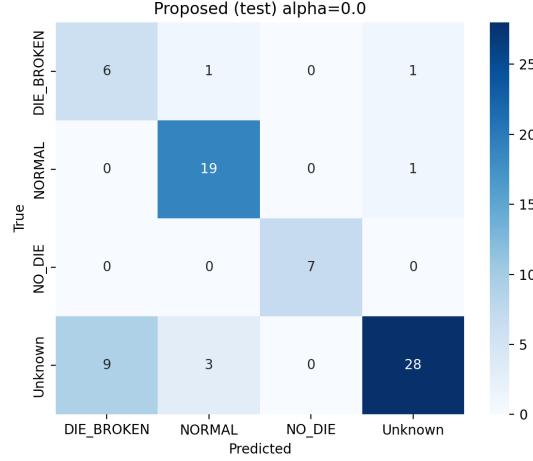


Figure 3: Confusion matrix of the proposed method with $\alpha = 0.2$. Most unseen defects are correctly rejected as Unknown while maintaining strong performance on known classes.

MSP improves unknown detection slightly, increasing macro F1 to 0.607. Energy-based detection achieves a macro F1 of 0.632, improving unknown recall compared with MSP but reducing known-class accuracy.

A substantially larger improvement is obtained with Mahalanobis distance. By measuring feature-space deviation from known-class centroids, Mahalanobis increases macro F1 to 0.790 while maintaining perfect unknown precision.

Combining Energy and Mahalanobis through a hard rejection rule yields a modest improvement to 0.801.

Finally, the proposed diagonal-covariance Energy–Mahalanobis fusion achieves the best overall performance, reaching a macro F1 score of 0.870. More importantly, the proposed method improves unknown recall from 0.625 to 0.900 compared with the hard Energy+Mahalanobis rule, while maintaining competitive known-class accuracy.

Compared with the closed-set baseline, the proposed method achieves

$$\frac{0.870 - 0.517}{0.517} = 68.4\%$$

relative improvement in macro F1.

As shown in Figure 3, the proposed method correctly rejects most unseen defect samples while preserving strong performance on known categories.

5 Discussion

The experimental results suggest that feature-space geometry is substantially more informative than classifier confidence for distinguishing unseen wafer defects. Mahalanobis distance outperforms both MSP and Energy, indicating that geometric deviation from known-class feature manifolds is a strong signal for open-set rejection. This explains why Mahalanobis distance substantially outperforms MSP and Energy.

The diagonal covariance approximation further improves robustness under limited data by reducing covariance estimation variance. Finally, Energy and Mahalanobis provide complementary signals: Energy captures predictive uncertainty, while Mahalanobis measures geometric deviation from known-class manifolds. Their fusion yields the strongest overall performance.

A notable observation is that the proposed method improves unknown recall substantially while sacrificing a portion of known-class accuracy. This trade-off is expected in open-set recognition

because aggressively rejecting uncertain samples inevitably increases false rejection of known samples. In practical semiconductor inspection systems, the preferred operating point depends on the relative cost of missed defects versus false alarms.

6 Conclusion

We investigated open-set wafer defect classification under a small-data setting. While a closed-set ResNet18 classifier achieved strong known-class performance, it failed to recognize unseen defects.

By combining diagonal-covariance Mahalanobis distance with Energy-based soft fusion, the proposed approach improved macro F1 from 0.517 to 0.870. These results demonstrate that substantial gains can be achieved through post-hoc open-set calibration without modifying the underlying classifier architecture. This finding suggests that simple post-hoc calibration techniques can provide a practical solution for real-world semiconductor inspection systems where unseen defects frequently occur.

References

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